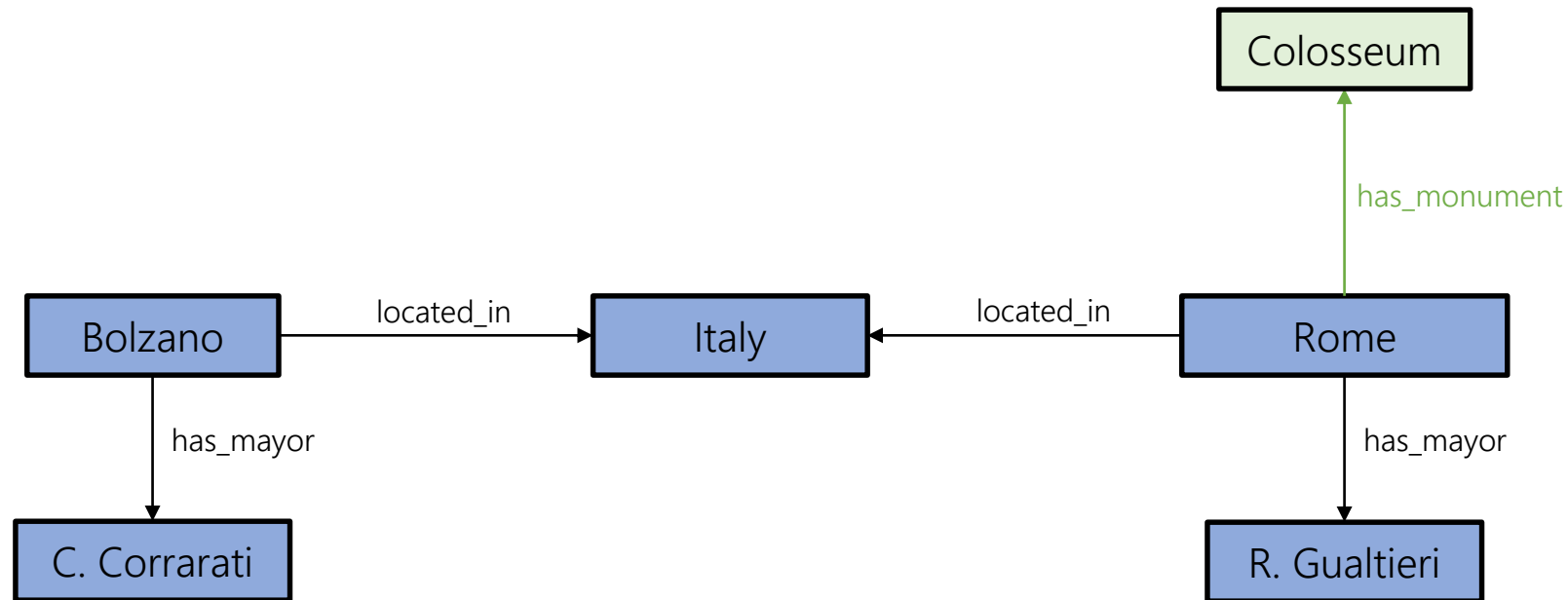


Continual Knowledge Graph Embedding

ISWC'26

Evolving Knowledge Graphs:

Real-world KGs are **not static**, as new triples, containing new entities and relations are incorporated.



Evolving Knowledge Graphs:

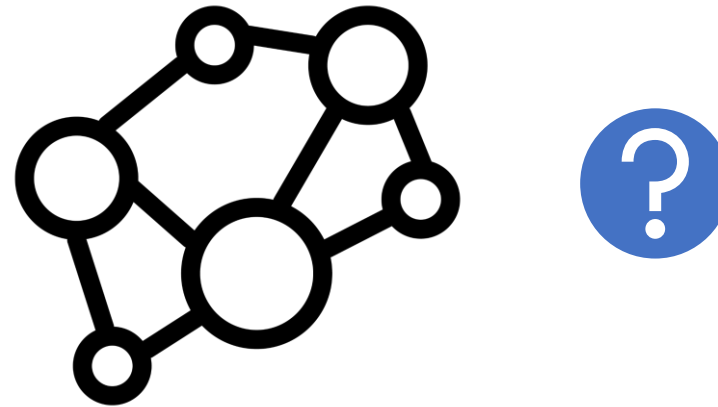
Real-world KGs are **not static**, as new triples, containing new entities and relations are incorporated.



From 40M entities in 2019 to 90M entities in 2021

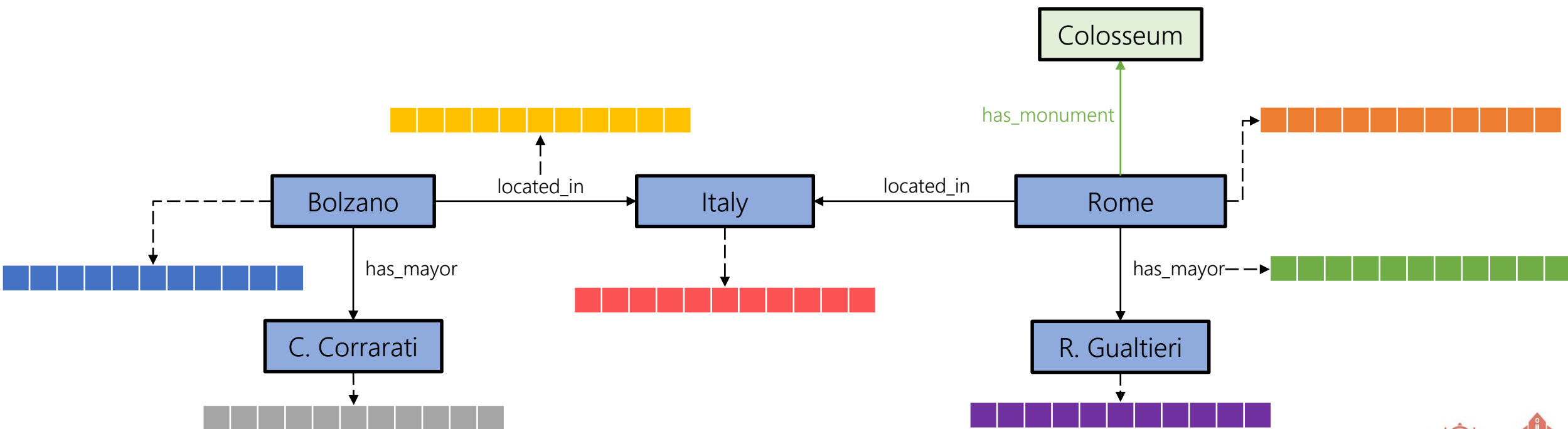
Exploiting Evolving Knowledge Graphs:

This new information can be directly exploited by **querying the graph**.



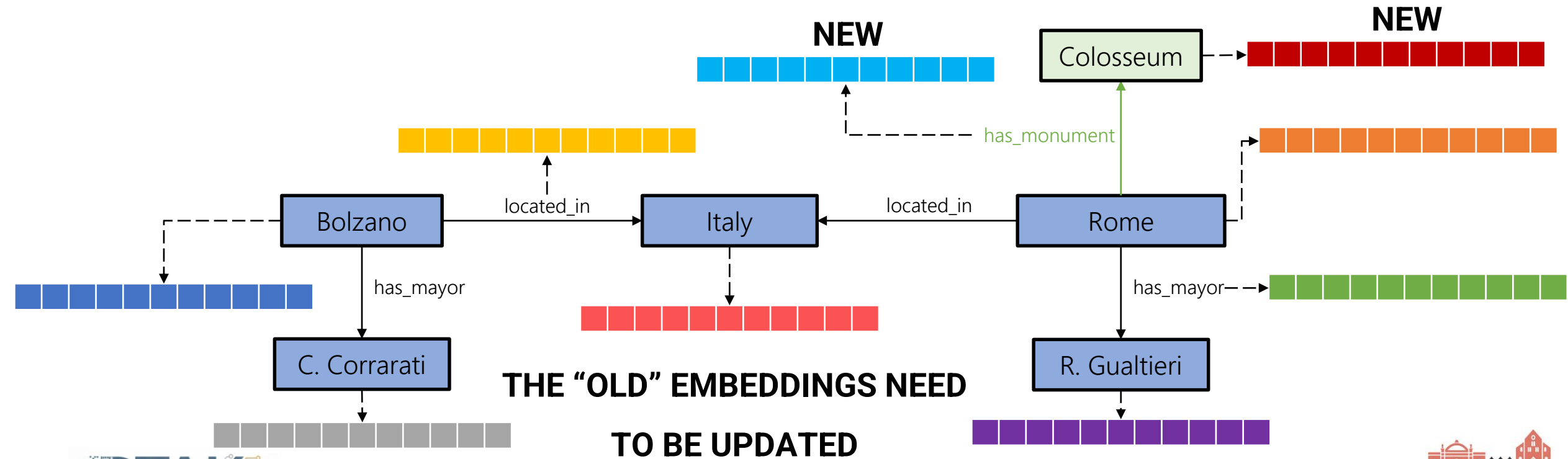
Exploiting Evolving Knowledge Graphs:

However, it is not straightforward for **KGEs**. If we want them to remain **useful** for downstream task and accurately represent the **actual state of the KG**, they need to be **maintained**.



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The KGE models that we saw in the previous section **assume a static KG**, so they alone are **not fit for evolving KGs**.

Adapting Knowledge Graph Embeddings:

However, it is not straightforward for **KGEs**. If we want them to remain **useful** for downstream task and accurately represent the **actual state of the KG**, they need to be **maintained**:

- We need to **learn embeddings** for the new entities and relations.
- We need to **update existing embeddings** to accommodate new information.

!

Adapting Knowledge Graph Embeddings:

Methods that **do not** take into account the **update of exiting embeddings** are known as **Inductive Knowledge Graph Embeddings**. They typically use **neighborhood aggregation** (e.g., with GNNs) or **feature encoding**.

They are useful in **certain domains** (e.g., we only need to obtain the embedding of a user to perform recommendations via link prediction) but fall out of the scope of this tutorial.

[1] Lee, Jaejun, Chanyoung Chung, and Joyce Jiyoung Whang. "Ingram: Inductive knowledge graph embedding via relation graphs." *International conference on machine learning*. PMLR, 2023.

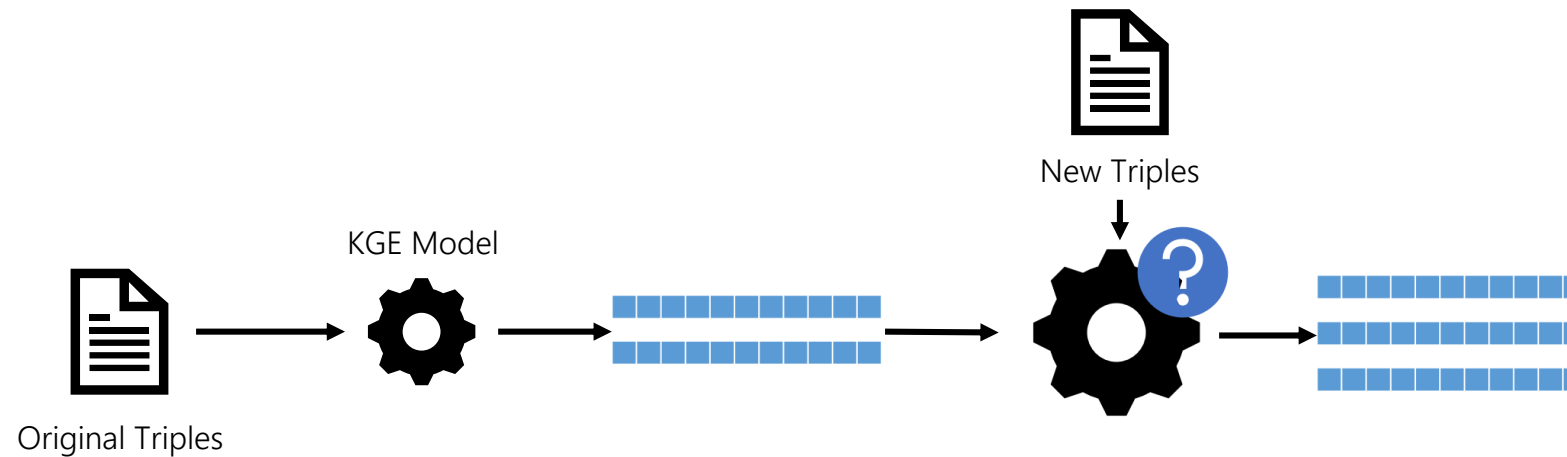
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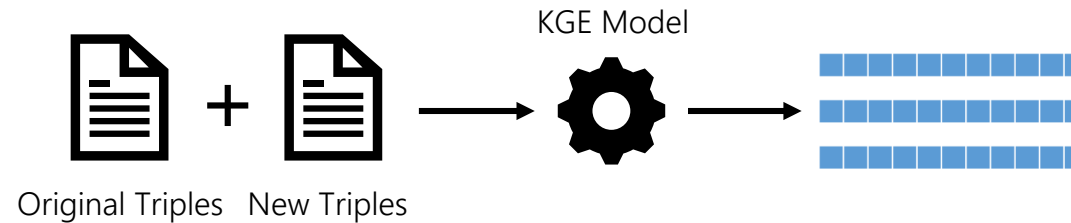


Adapting Knowledge Graph Embeddings:



Adapting Knowledge Graph Embeddings:

The most straightforward solution is **retraining**. This is, using all the data to obtain again the KGE **from scratch**.



PROS:

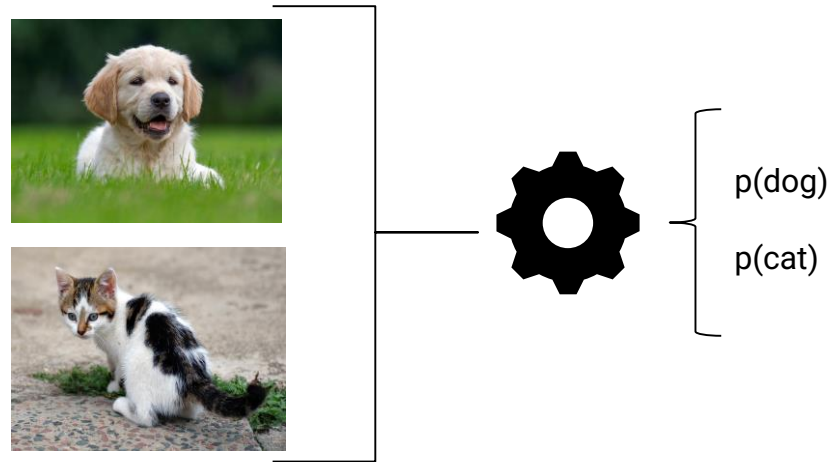
- **Good representations** are obtained.

CONS

- **Expensive** and **time-consuming**, specially in highly dynamic environments.
- Need to **update downstream tasks**, as the generated **embedding space** is not the same.

Continual Learning:

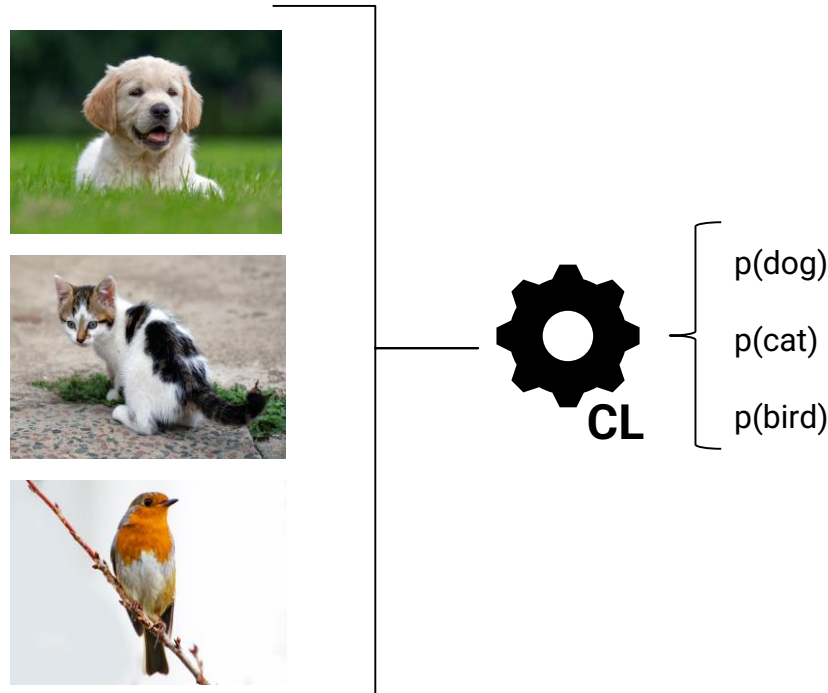
To avoid training from scratch, the **Continual Learning** concept has emerged. It is described as ability of system to continuously **acquire**, **adapt**, and **transfer** knowledge from a stream of data while **preserving previously learned knowledge** (i.e., avoiding **catastrophic forgetting**).



Continual Learning:

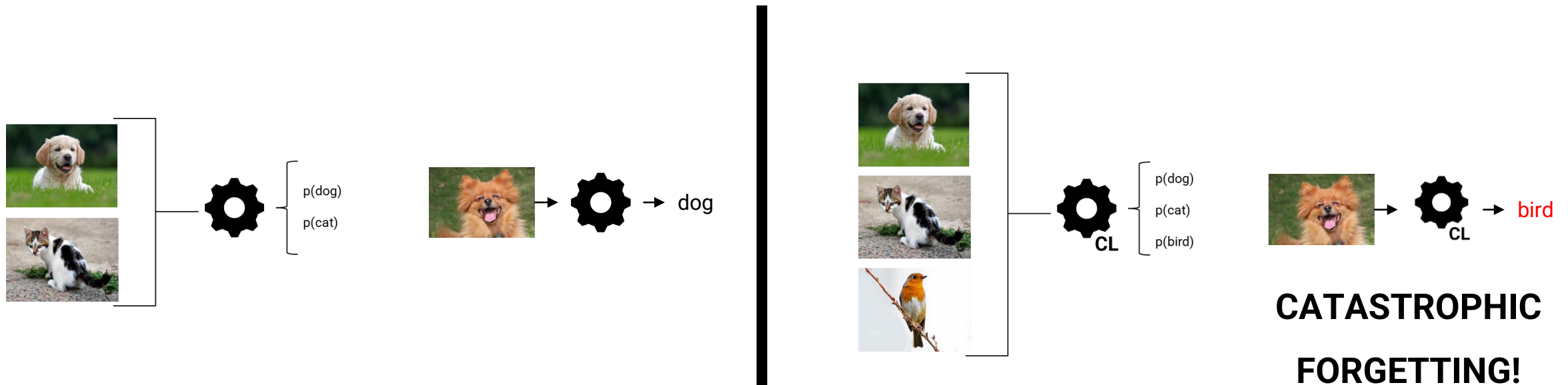
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Class Incremental Learning



Continual Learning:

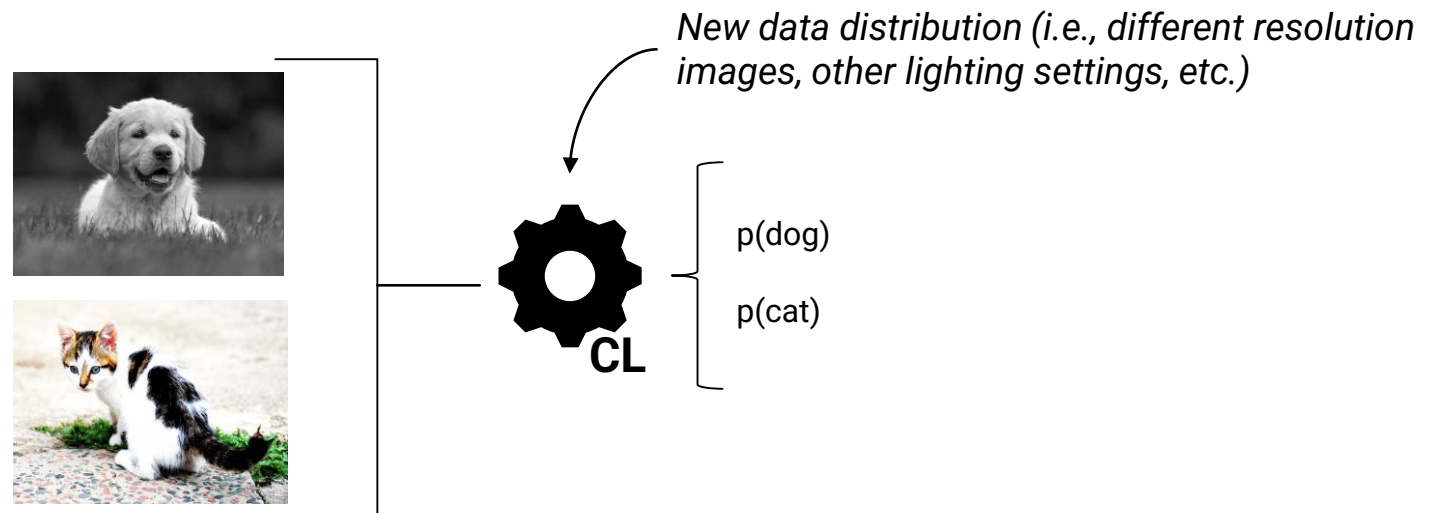
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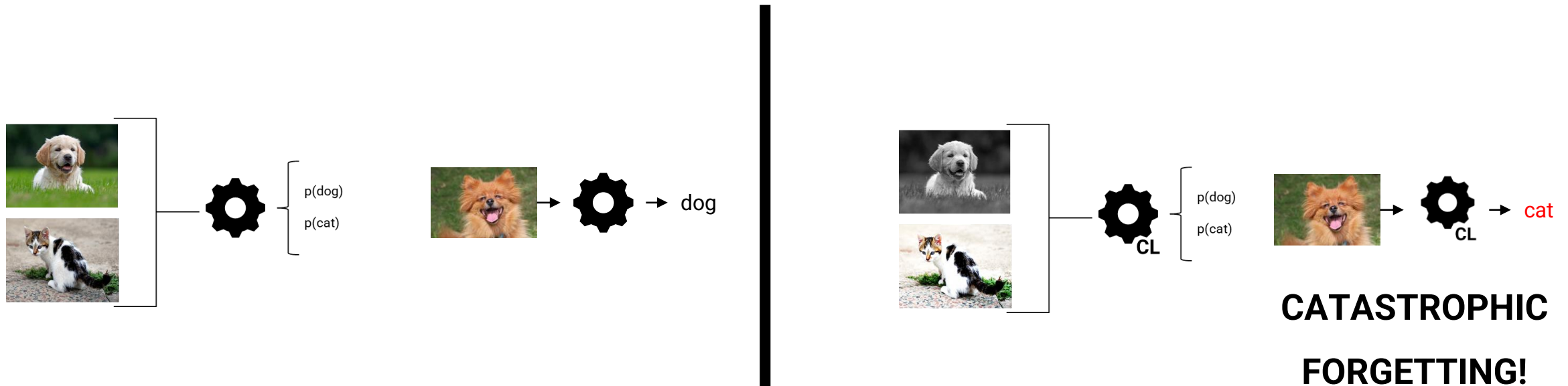
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Domain Incremental Learning



Continual Learning:

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Continual Learning:

To avoid training from scratch, the **Continual Learning** concept has emerged. It is described as ability of system to continuously **acquire**, **adapt**, and **transfer** knowledge from a stream of data while **preserving previously learned knowledge** (i.e., avoiding **catastrophic forgetting**).

In **KGEs**, this has given rise to the field of **Continual Knowledge Graph Embedding**.

Continual Knowledge Graph Embedding:

Therefore, in CKGE we would like learn new embeddings and incorporate new information to the existing ones **while not losing the already acquired knowledge**.

(Bari, located_in, ?)

BEFORE

(Bari, located_in, Italy)

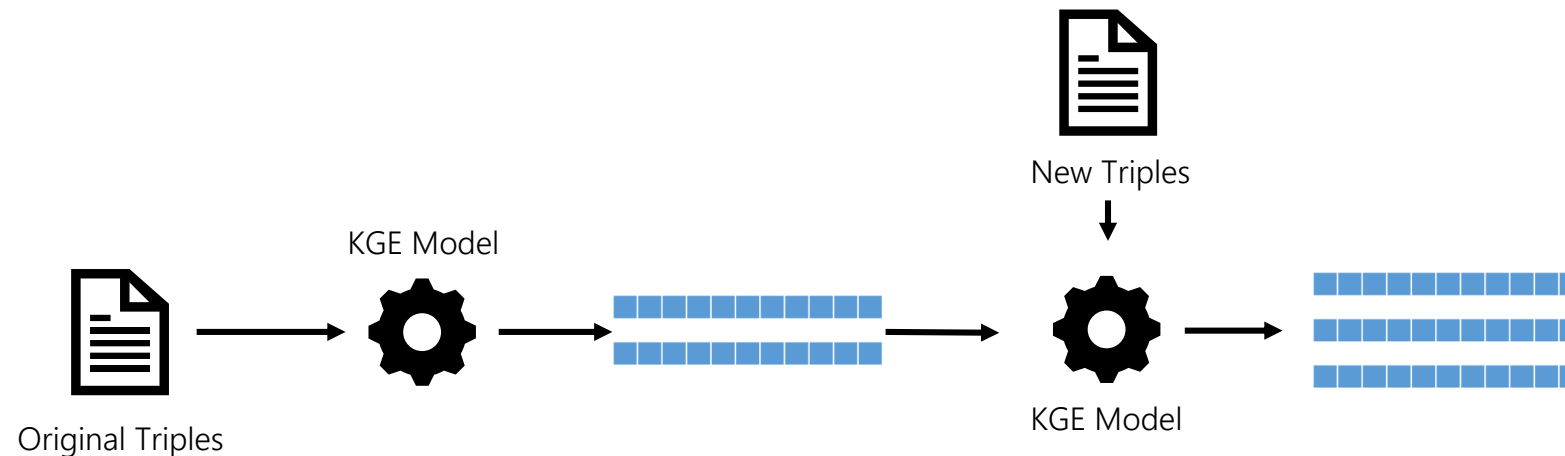
AFTER

(Bari, located_in, France)

**CATASTROPHIC
FORGETTING!**

Continual Knowledge Graph Embedding:

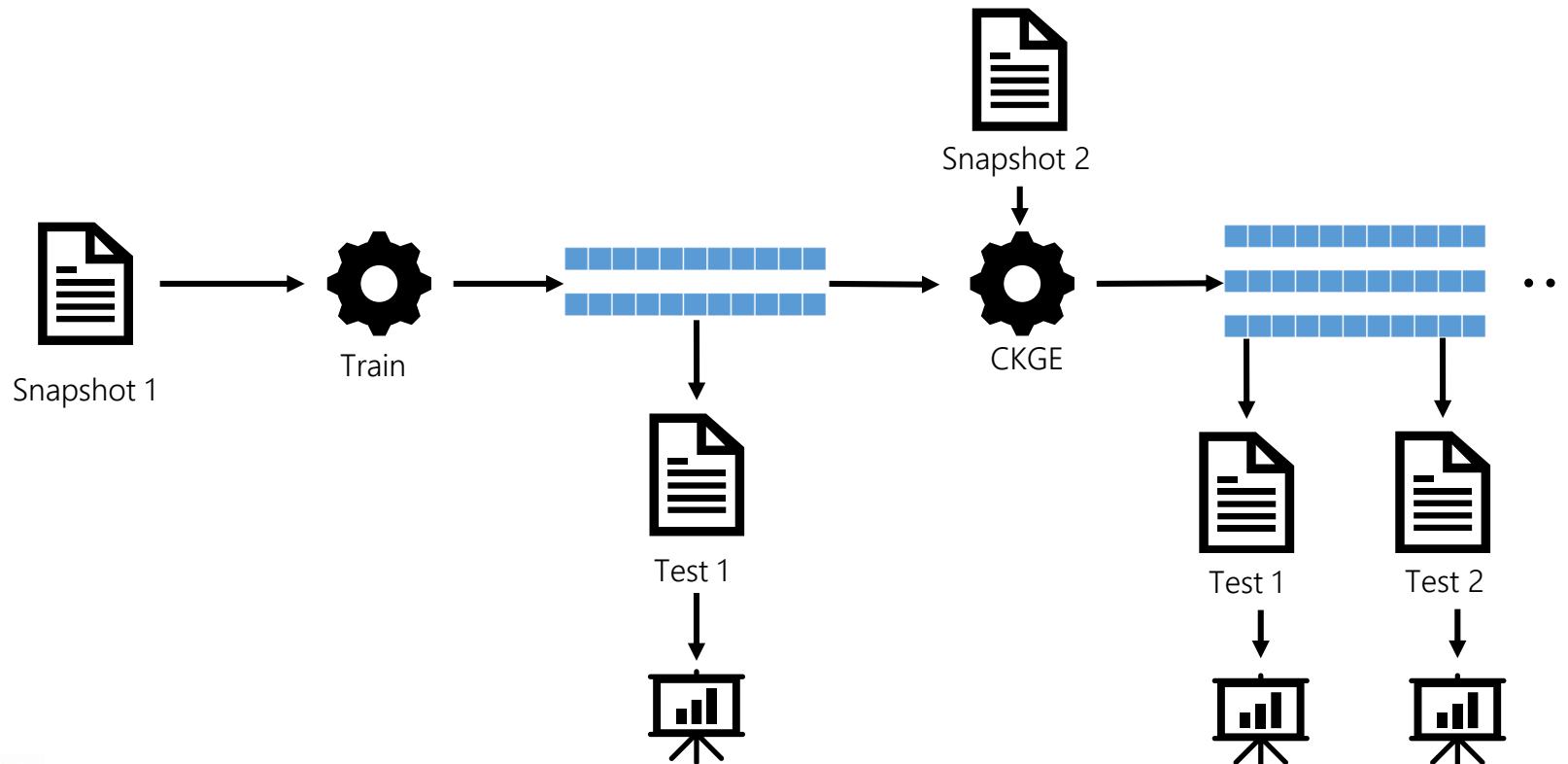
The, most naïve CKGE strategy is **fine-tuning**, which involves training over the existing embeddings with the new data. This is a **simple** and **inexpensive** approach, but experiences **large amounts of catastrophic forgetting**.



In the next session, we will explore the **state-of-the-art CKGE approaches**.

Evaluating CKGE:

To evaluate CKGE, the previously introduced metrics **MRR** and **Hits@k** are also used, but now are **weighted** by the size of each test set.



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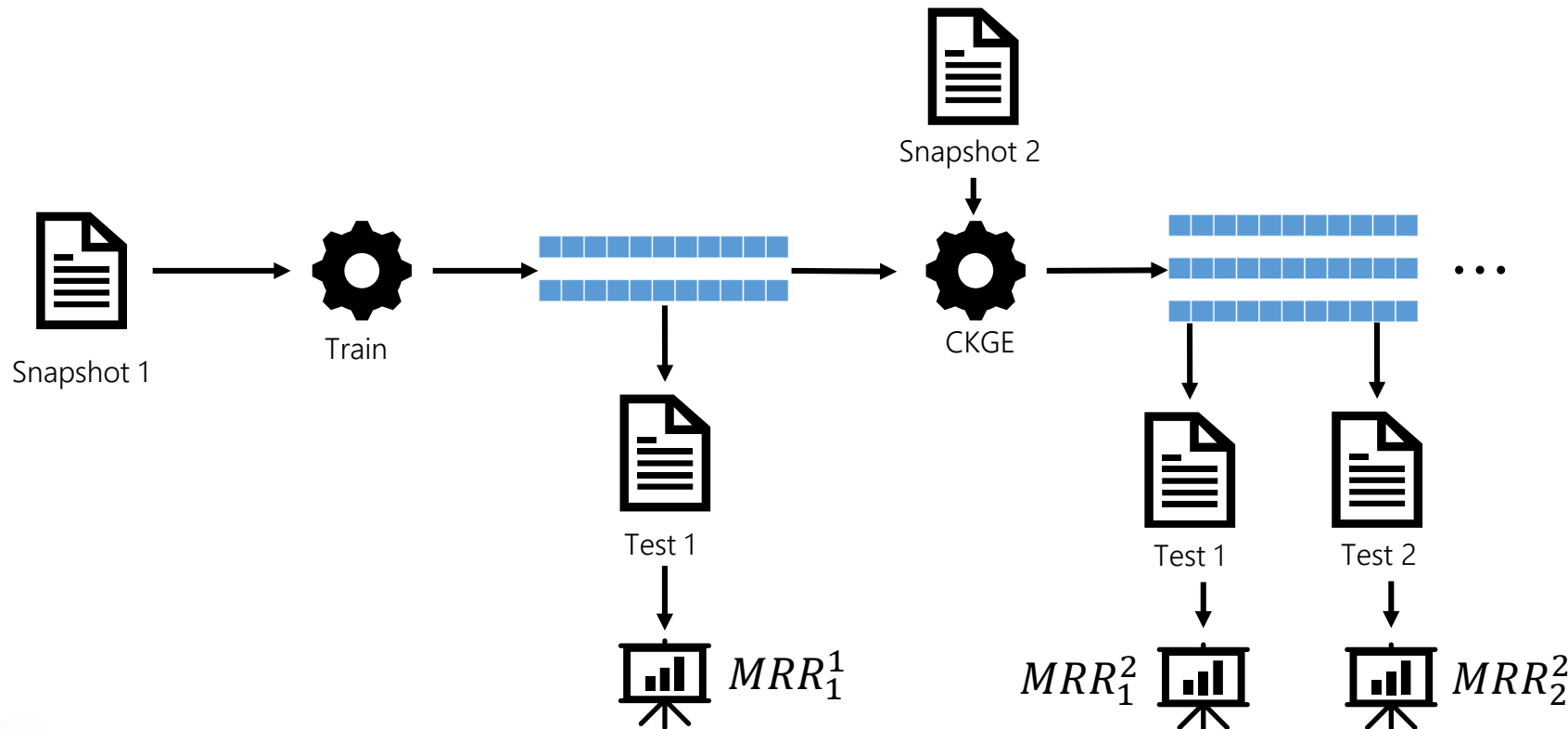
$$MRR_{CL} = \sum_{i=1}^N \frac{|T_i|}{\sum_{j=1}^N |T_j|} MRR_i^N$$

MRR_i^N

MRR of the test set **i** after
training in snapshot **N**.

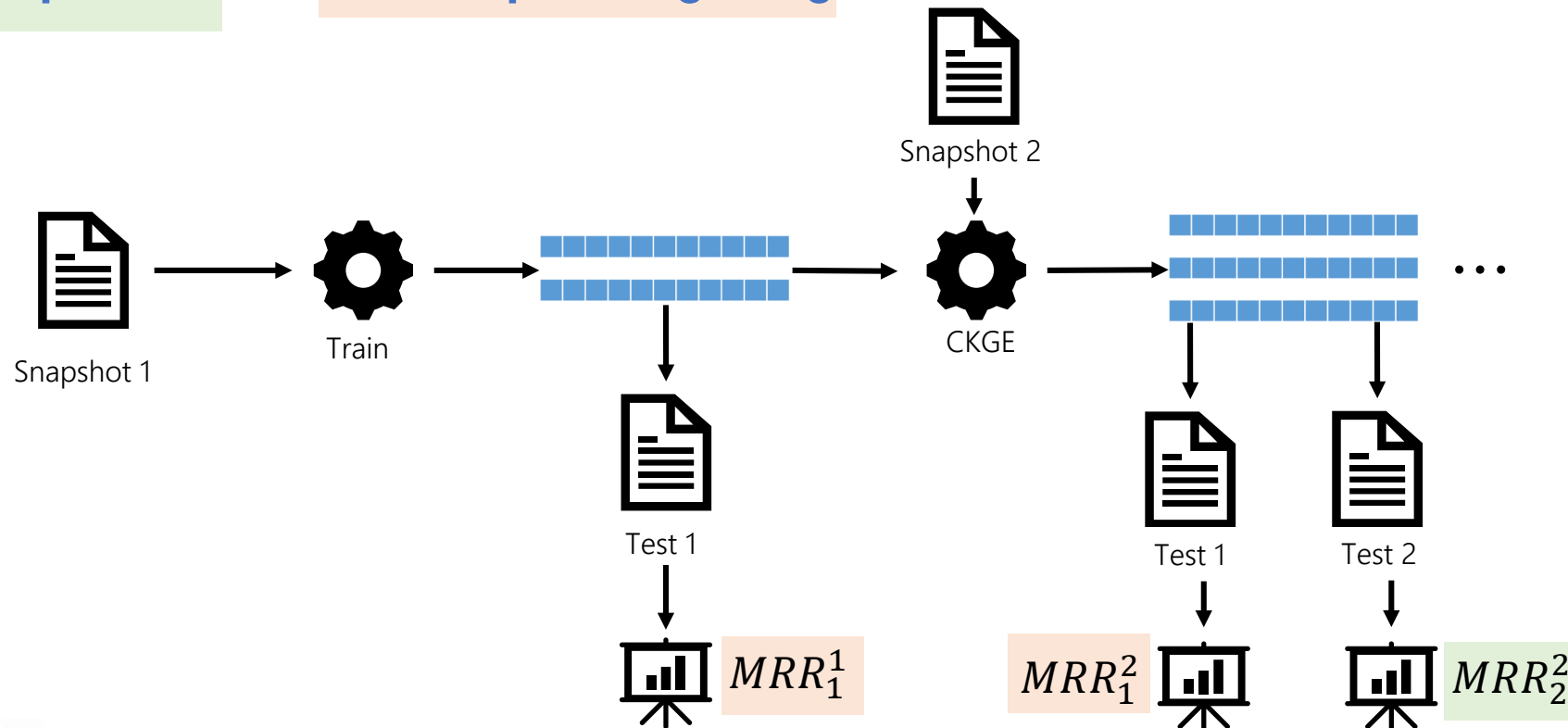
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$$\Omega_{new} = \sum_{i=2}^N MRR_i^i \frac{|T_i|}{\sum_{j=2}^N |T_j|}$$

Evaluating CKGE:

Additionally, we concretely assess the desired Continual Learning behavior by measuring **knowledge acquisition** and **catastrophic forgetting**.

$$BWT = \frac{1}{N-1} \sum_{i=1}^{N-1} (MRR_N^i - MRR_i^i)$$

$$\Omega_{base} = \sum_{i=1}^{N-1} \frac{MRR_N^i}{MRR_i^i}$$

- *Same contribution regardless of the test dataset size.*
- *Same contribution regardless of the incremental steps the data has undergone*

Evaluating CKGE:

Additionally, we concretely assess the desired Continual Learning behavior by measuring **knowledge acquisition** and **catastrophic forgetting**.

$$ACF = \sum_{i=1}^{N-1} \frac{(MRR_N^i - MRR_i^i)}{(N - i)MRR_i^i} \frac{|T_i|}{\sum_{j=1}^{N-1} |T_j|}$$

Evaluating CKGE: Benchmark Datasets

The **SOTA benchmark datasets** [2,3] consist **of 5 snapshots** (1 **base** snapshot and 4 **incremental** snapshots) build from **FB15K-237** [4].

[2] Y. Cui, Y. Wang, Z. Sun, W. Liu, Y. Jiang, K. Han, W. Hu, *Lifelong embedding learning and transfer for growing knowledge graphs*, in: *AAAI*, 2023.

[3] J. Liu, W. Ke, P. Wang, Z. Shang, J. Gao, G. Li, K. Ji, Y. Liu, *Towards continual knowledge graph embedding via incremental distillation*, in: *AAAI*, 2024.

[4] K. Toutanova, D. Chen, *Observed versus latent features for knowledge base and text inference*, in: *Proceedings of the 3rd workshop on continuous vector space models and their compositionality*, 2015, pp. 57–66.

Evaluating CKGE: Benchmark Datasets

The **SOTA benchmark datasets** [2,3] consist **of 5 snapshots** (1 **base** snapshot and 4 **incremental** snapshots) build from **FB15K-237** [4]:

- ENTITY: each snapshot has the same number of new entities
- RELATION: each snapshot has the same number of new relations
- FACT: each snapshot has the same number of new facts
- HYBRID: uniform expansion of entities, relations and facts
- GraphEqual: each snapshot has the same number of facts (allows 2 new entities per triple)
- GraphHigher: the number of facts increases with the snapshots
- GraphLower: the number of facts decreases with the snapshots

Questions?